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ORIGINAL ARTICLE

Prevalence of Visual Impairment and Utilization of Rehabilitation Services in the Visually Impaired Elderly Population of Quebec

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ABSTRACT: Data on the prevalence of reported visual impairment and on the utilization of rehabilitation services were collected on a sample consisting of 1777 community-residing people aged at least 65 years. A visual disability was considered to be present if the answer to at least one of the following two questions was positive: Do you have trouble reading ordinary newsprint with glasses (if normally worn)? Do you have trouble clearly seeing the face of someone 12 feet away with glasses (if normally worn)? Prevalence of a reported near disability was 7.6%, prevalence of a reported distance disability was 4.4%, and 3.5% of subjects reported both types of disability. In a subsample of the surveyed population, the positive predictive value was 21% and the negative predictive value was 100%, using moderate or worse visual impairment as the gold standard. Among those answering yes to both questions, 11.4% received services from a rehabilitation center and 10.0% from a nonprofit agency. The utilization rates (adjusted to apply only to those whose visual impairment was confirmed by visual examination) reached 20% for rehabilitation centers and 17.5% for nonprofit agencies. Low utilization of rehabilitation services raises questions concerning the role of general eye care practitioners, community-based health centers, and rehabilitation centers in the rehabilitative process of the visually impaired elderly. (Optom Vis Sci 2002;79:416–423)

Key Words: prevalence, visual impairment, rehabilitation services, elderly population

Information on the prevalence of blindness and visual impairment is critical for quantifying the need for care and rehabilitation services for the visually disabled. Population-based studies of visual impairment and ocular disorders are the most valid method for determining prevalence. Unfortunately, oculo-visual assessments of large general populations are very costly and time consuming. As an alternative to examination studies, national agencies such as the National Center for Health Statistics in the United States and Statistics Canada regularly use questionnaire-based surveys of large samples to gather statistics on health and disability status.[1–3] Recent data on reported visual impairment published by the National Center for Health Statistics and Statistics Canada, together with those from examination studies on clinically assessed visual impairment, indicate that most individuals identified as visually disabled are elderly.[1–5] The elderly subgroup of the visually impaired population has been identified as being underserved.[6]

Visually impaired elderly may receive rehabilitation services from different and sometimes multiple sources. In Canada, these sources include rehabilitation centers, private and hospital-based low-vision clinics, and nonprofit agencies.[7] In the province of Quebec, rehabilitation centers provide a full complement of services funded by the provincial Ministry of Health and Social Services. These centers were established to serve visually impaired individuals and are geographically located to meet the needs of the population. Rehabilitation services are provided free of charge for all individuals who meet the International Classification of Diseases (9th revision) definition of visual handicap,[8] whereas rehabilitative aids may be loaned or purchased at low cost.[9] In Quebec, nonprofit agencies, such as the Canadian Institute for the Blind, are mainly involved in advocacy activities, job placement, and cultural and leisure activities and, in contrast to most of the other provinces of Canada, not at all in low-vision and rehabilitation services. The private or hospital-based low-vision clinics offer clinical assessment but not the full scope of rehabilitative services. Prescription of low-vision devices is offered, but patients must pay for the required visual aids. The International Classification of Diseases (9th revision) criteria are also used by the nonprofit agen-

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cies, but other low-vision clinics may have less stringent access criteria. Even though utilization by the elderly of rehabilitation services for the visually impaired has risen during the last decade in Canada, the proportion of users is still very low compared with the prevalence of individuals reporting visual disability estimated from Statistics Canada surveys.[10, 11] Furthermore, the proportion of visually impaired elderly who use rehabilitation services is not well documented.[6, 12, 13] To determine what proportion of visually impaired elderly use rehabilitation services, a two-stage research project was implemented. The purpose of the first stage was to collect data on the prevalence of reported visual disability and on the utilization of rehabilitation services in a large sample of community-residing people aged at least 65 years. The purpose of the second stage was to estimate the sensitivity, specificity, and predictive values of two survey questions using the results of a visual examination as the gold standard.

METHODS

The first stage of the study was part of a large survey on health and aging in Canada, the Canadian Study of Health and Aging (CSHA), which took place from 1991 to 1992. Study methods for the CSHA have been described elsewhere.[14] Briefly, sampling areas were identified in the vicinity of each of the 18 research centers participating in the CSHA. A random sample of individuals aged at least 65 years was drawn for each selected area, with predetermined sampling fractions for age and sex categories. Individuals aged 75 to 84 years and aged 85 years or older were oversampled.

There were four research centers in Quebec. The community sample in Quebec was made up of 1800 persons aged at least 65 years. Data on visual impairment were collected during a home interview carried out as part of the CSHA study. As defined in previous Statistics Canada surveys such as the Health and Activity Limitations Survey (HALS),[2] a visual disability was considered to be present if the answer to at least one of the following two questions was positive:

- Do you have trouble reading ordinary newsprint with glasses or contact lenses if normally worn?
- Do you have trouble seeing clearly the face of someone 12 feet away with glasses or contact lenses if normally worn?

If the answer was positive to at least one of these two questions, subjects were asked whether they received services for visual impairment from either a rehabilitation center, a low-vision clinic other than in a rehabilitation center (e.g., in a hospital or in the private office of an eye care practitioner), or a nonprofit agency that serves the visually impaired (e.g., Canadian National Institute for the Blind). Prevalence and 95% confidence intervals were computed for each sex and age category and for the whole study population, taking into account the sampling design.[15] Whenever the conditions for computation of the usual confidence interval for a proportion were not respected, a binomial 95% confidence interval for small samples was used.[16] Weighted prevalences were calculated to correct for the discrepancy between the proportion of subjects included in the study sample in the different sex and age categories and the corresponding proportion of people in the population of Quebec.

In the second stage of the study, we assessed the validity of the questionnaire in a subsample of our study group. Among subjects in the Montreal area not eligible for other components of the CSHA, all positive subjects, along with a random sample of negative subjects, were selected to undergo a visual examination at home. Subjects were considered positive if they answered "yes" to at least one of the two screening questions. The examination included measurement of visual acuity (VA), visual fields, retinoscopy, subjective refraction, aplanation tonometry, and funduscopy. Distance VA was measured at 4 m using the Early Treatment Diabetic Retinopathy Survey charts[17] in a back-lighted box.[18] Near VA was measured at 0.4 m with a modified near Early Treatment Diabetic Retinopathy Survey chart. Visual fields were assessed with a 0.333-m radius portable arc-perimeter and a 3-mm white index. VA was measured twice: first with the usual correction, if any, worn by the subject (presenting VA)[5] and again with the best correction as determined by refraction (best-corrected VA). The first author (JG), an optometrist who was not aware of the subjects' questionnaire responses at the time of the examination, carried out the visual assessment.

Subjects were classified according to the VA and visual field in their best eye into the International Classification of Diseases (9th revision) categories of visual impairment.[8] Individuals with a best VA between 6/12 and 6/18 were considered to have a mild visual impairment,[5] individuals with a best VA between 6/21 and 6/54 were considered to have a moderate visual impairment, and subjects having a VA $\leq$ 6/60 or a visual field $< 20°$ were considered to have a severe visual impairment.[8] The classification was done twice: the definition using the presenting VA is referred to as the "functional visual impairment,"[19] and the other, with the best-corrected VA, is referred to as the "visual impairment." Using the resulting classification as the diagnostic standard and the results of the questionnaire as the test to be validated, the subjects were reclassified as true positives (a), false positives (b), false negatives (c), and true negatives (d). Crude sensitivity, $Sn = a/(a + c)$, and crude specificity, $Sp = d/(b + d)$, of the questionnaire were computed, and their 95% confidence intervals (95% CI) were determined from the binomial 95% CI tables for small samples.[20] Because the sampling fraction for subjects with a positive questionnaire result was different from the sampling fraction for subjects with a negative questionnaire result, the sensitivity and specificity estimated in this study cannot be applied to an unselected population. The positive predictive value, $PPV = a/(a + b)$, and the negative predictive value, $NPV = d/(c + d)$, were also computed. The adjusted prevalence (Pr) of visual impairment was computed according to the formula proposed by Choi[21]: $Pr = [(m_1)PPV] + [(m_0)(1 - NPV)]/(m_1 + m_0)$, where m_1 is the number with positive tests and m_0 the number with negative tests in the unselected population. Results were computed using four different diagnostic criteria and two different test criteria.

Diagnostic Criteria

- Mild functional impairment or worse (presenting near or distance VA $\leq$ 6/12) versus no functional impairment (presenting near or distance VA $>$ 6/12).
- Moderate functional impairment or worse (presenting near or distance VA $\leq$ 6/21) versus no functional impairment or mild

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functional impairment (presenting near or distance VA > 6/21).
- Mild impairment or worse (best-corrected near or distance VA ≤ 6/12) versus no impairment (best-corrected near or distance VA > 6/12).
- Moderate impairment or worse (best-corrected near or distance VA ≤ 6/21) versus no impairment or mild impairment (best-corrected near or distance VA > 6/21).

Test Criteria

- A positive response to the near-vision task question versus a negative response.
- A positive response to the far distance-vision question versus a negative response.
- At least one positive response versus two negative responses.
- Two positive responses versus at least one negative response.

The ethics committee of the Center Hospitalier Côte-des-Neiges approved this research protocol.

RESULTS

In the first stage, 1777 individuals (98.7% of the eligible subjects), including 1106 women and 671 men, completed the questionnaire on reported visual disability. The prevalence of disability, according to the HALS Statistics Canada survey questions,[2] is shown in Table 1. For both near- and distance-vision complaints, the prevalence rose with age. The prevalence was higher in women relative to men in the younger age groups and higher in men than in women for the older age groups for both near and distance visual impairment, although this difference was not statistically significant based on an examination of the respective confidence intervals. The prevalence in the whole population of at least 65 years of age, when weighted to remove the effect of the differing sampling fractions, was 7.6% for near-vision impairment and 4.4% for impairment in distance vision. The prevalence of reported near-vision impairment was significantly higher than the prevalence of reported impairment for distance vision. When considering the prevalence of reported visual impairment according to the number of positive responses (Table 2), similar trends were observed in prevalence according to age and sex categories. After adjustment to remove the effect of the differing sampling fractions, the prevalence in the whole population of at least 65 years of age was 8.5% according to the Statistics Canada criteria and 3.5% for both near- and distance-vision impairment. These two proportions represent, respectively, about 60,100 and 24,800 individuals in the Quebec population. Table 3 shows the use of services by elderly persons aged at least 65 years who reported visual impairment. The services of private and hospital-based low-vision clinics were the most frequently used.

Among the 144 subjects eligible for the second stage of the study, 31 were lost to follow-up and 42 refused to participate. We were unable to schedule an eye examination before the end of the study for six of the remaining 71 subjects. Among the 65 subjects who received the visual examination, 19 had at least one positive response to the questionnaire, and 46 had two negative responses. The final participation rate was 45.1%, 40.4% among those with a positive questionnaire and 47.4% among those with a negative questionnaire. Participating and nonparticipating subjects did not differ with respect to age or visual disability status. However, the proportion of nonparticipants was larger among women than men (62% versus 41%, respectively). After refraction, only two subjects changed from mild visual impairment to no visual impairment. Hence, the results presented in the following tables are restricted to only functional impairment. The prevalence of mild functional near visual impairment or worse was 30.8%, whereas the prevalence of moderate functional near visual impairment or worse was 6.2%. The prevalence of mild functional distance visual impair-

TABLE 1.
Prevalence (%) of reported near- and far-vision impairment by sex and age.

Type of Reported Disability	Age (yr)	N	Female (N = 1106) (95% CI)	Male (N = 671) (95% CI)	Both Sexes[a] (N = 1777) (95% CI)
Near-vision task impairment	65 to 74	849	5.3 [3.3–7.3]	2.8 [1.1–4.5]	4.2 [2.8–5.6]
	75 to 84	736	14.1 [11.0–17.2]	10.4 [6.7–14.1]	12.7 [10.3–15.1]
	85 and over	192	20.1 [13.4–26.8]	28.3 [16.2–40.4]	22.6 [16.7–28.5]
	All 65 and over[a]		8.8 [7.2–10.4]	5.8 [4.2–7.4]	7.6 [6.4–8.8]
Distance-vision task impairment	65 to 74	849	3.7 [2.0–5.4]	1.7 [0.4–3.0]	2.9 [1.8–4.0]
	75 to 84	736	6.7 [4.5–8.9]	5.0 [2.4–7.6]	6.1 [4.4–7.8]
	85 and over	192	13.7 [8.0–19.4]	18.9 [8.4–29.4]	15.3 [10.2–20.4]
	All 65 and over[a]		5.2 [3.9–6.5]	3.2 [2.0–4.4]	4.4 [3.5–5.3]

[a] Weighted to remove the effect of the different sampling fractions.

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TABLE 2.
Prevalence (%) of reported visual impairment according to number of positive questionnaire responses, by sex and age.

Age (yr)	Female		Male		Both Sexes[a]	
	At Least One Positive Response (N = 133) (95% CI)	Two Positive Responses (N = 57) (95% CI)	At Least One Positive Response (N = 58) (95% CI)	Two Positive Responses (N = 23) (95% CI)	At Least One Positive Response (N = 191) (95% CI)	Two Positive Responses (N = 80) (95% CI)
65 to 74	5.9 [3.8–8.0]	3.1 [1.6–4.6]	3.4 [1.5–5.3]	1.1 [0.2–4.3][b]	4.8 [3.4–6.2]	2.2 [1.2–3.2]
75 to 84	15.3 [12.1–18.5]	5.5 [3.5–7.5]	11.9 [8.0–15.8]	3.5 [1.3–5.7]	14.0 [11.5–16.5]	4.7 [3.2–6.2]
85 and over	22.3 [15.4–29.2]	11.5 [6.2–16.8]	28.3 [16.2–20.4]	18.9 [8.4–29.4]	24.1 [18.0–30.2]	13.7 [8.8–18.6]
All 65 and over[a]	9.7 [8.0–11.4]	4.3 [3.1–5.5]	6.6 [4.9–8.3]	2.4 [1.4–3.4]	8.5 [7.3–9.7]	3.5 [2.7–4.3]

[a] Weighted to remove the effect of the different sampling fractions.
[b] This 95% confidence interval was taken from the binomial 95% confidence intervals tables for small samples.[20]

TABLE 3.
Utilization (%) of rehabilitation services among elderly persons aged at least 65 years who report visual disability (weighted to remove the effect of the different sampling fractions in the various age and sex categories).

Source of Services	Visually Impaired According to the Question on Near-Vision Task (95% CI)	Visually Impaired According to the Question on Far-Vision Task (95% CI)	Visually Impaired According to at Least One Positive Response (95% CI)	Visually Impaired According to Only One Positive Response (95% CI)	Visually Impaired According to Both Positive Responses (95% CI)
Rehabilitation centers	5.4 [1.8–9.0]	8.5 [2.7–14.3]	5.2 [1.7–8.7]	0.8 [0.0–5.4][a]	11.4 [3.7–19.1]
Low-vision clinics in hospitals and private offices	22.9 [16.3–29.5]	22.0 [13.2–30.8]	24.0 [17.3–30.7]	24.6 [15.9–33.3]	23.3 [13.0–33.6]
Nonprofit agencies	4.8 [1.3–8.3]	8.2 [2.3–14.1]	4.2 [0.9–7.5]	0 [0.0–3.6][a]	10.0 [2.5–17.5]
Any service	24.9 [18.1–31.7]	25.5 [16.5–34.5]	26.0 [19.2–32.8]	24.6 [15.9–33.3]	28.1 [17.4–38.8]

[a] These 95% confidence intervals were taken from the binomial 95% confidence intervals tables for small samples.[20]

ment or worse was 18.5%, whereas the prevalence of moderate functional distance visual impairment or worse was 6.2% (data not tabulated).

Estimates of crude sensitivity and specificity and as predictive values of the questionnaire using different criteria are presented in Table 4. Due to the small number of true positives, the precision of the crude sensitivity and of the PPV was very low. Nevertheless, when these figures were used to adjust the prevalence among persons aged at least 65 years in the Quebec population, the prevalence of mild functional impairment or worse was 9.5%, whereas the prevalence of moderate functional impairment or worse was 2.6%.[21]

To be eligible for the services offered at rehabilitation centers in Quebec, individuals must have a distance VA $\leq$ 6/21 (moderate impairment or worse). The PPV of the questionnaire indicates that only 57.0% of those claiming disabilities at both near and far distance had moderate visual impairment or worse and were eligible for rehabilitation services. After adjusting the number of potential users by a factor of 0.57,[21] the estimated utilization rate of rehabilitation services reached 20.0% for the rehabilitation centers, 17.5% for the nonprofit agencies, and 49.3% for the other low-vision clinics.

DISCUSSION

In this study, the prevalence of reported visual impairment based on a criterion of at least one positive questionnaire response and the pattern of increase with age were similar to those reported in the HALS for 1986[3, 4] and 1991[22] and, more particularly, to the prevalence specific for the province of Quebec[23] but not with those of the 1998 Quebec survey.[24] Our results are also very similar to those derived from the National Center for Health Statistics by Nelson.[1] Fig. 1 shows the age-specific prevalences of this and three other questionnaire-based surveys (reported visual disability)[1, 23, 24] and of four population-based studies (clinically assessed visual impairment).[5, 19, 25, 26] The study by Salive et al.[19] used a

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TABLE 4.
Crude sensitivity, crude specificity, and positive and negative predictive values (%) of the questionnaire.[a]

	Functional Impairment (Presenting Visual Acuity)	
	Mild or Worse (VA ≤ 6/12)	Moderate or Worse (VA ≤ 6/21)
Positive response to the reading disability question vs. negative response to the reading disability question		
Crude sensitivity	75	100
(95% CI)	(43–95)	(40–100)
Crude specificity	81	75
(95% CI)	(69–91)	(62–85)
PPV	47	21
(95% CI)	(24–71)	(6–46)
NPV	93	100
(95% CI)	(83–99)	(92–100)
Positive response to the distance-vision disability question vs. negative response to the distance-vision disability question		
Crude sensitivity	50	100
(95% CI)	(21–79)	(40–100)
Crude specificity	98	95
(95% CI)	(90–100)	(88–99)
PPV	86	57
(95% CI)	(42–100)	(18–90)
NPV	90	100
(95% CI)	(81–97)	(94–100)
At least one positive response versus two negative responses		
Crude sensitivity	83	100
(95% CI)	(52–98)	(40–100)
Crude specificity	83	75
(95% CI)	(71–92)	(62–85)
PPV	53	21
(95% CI)	(29–76)	(6–46)
NPV	96	100
(95% CI)	(85–99)	(92–100)
Two positive responses versus at least one negative response		
Crude sensitivity	50	100
(95% CI)	(21–79)	(40–100)
Crude specificity	98	95
(95% CI)	(87–99)	(88–99)
PPV	86	57
(95% CI)	(42–96)	(18–90)
NPV	90	100
(95% CI)	(80–96)	(94–100)

[a] All these 95% confidence intervals were taken from the binomial 95% confidence intervals tables for small samples.[20]

case definition of moderate functional visual impairment and worse (presenting VA ≤ 6/21), whereas that by Tielsch et al.[5] and Klein et al.[25] used a definition of moderate visual impairment and worse (best-corrected VA ≤ 6/21) The more recent population-based study by Munoz et al.[26] used a less stringent definition of mild and worse visual impairment (best-corrected VA ≤ 6/12). For the studies conducted before 1992,[1, 5, 19, 23, 25] questionnaire-based survey results[1, 23] are in fairly close agreement with each other and with the present study; prevalence of reported visual disability is lower than the prevalence of visual impairment derived from population-based studies when visual impairment is defined by best-corrected VA[21, 25]and similar when defined by presenting acuity.[19] The two more recent studies suggest a reduction in the prevalence of both visual impairment and disability in the late 1990s,[24, 26] for all age categories for visual impairment[26], and at least in the younger ones for visual disability.[24] It may be that people answering visual disability questionnaires are reporting problems in regard to their presenting VA. Consequently, misclassification of visually impaired individuals by survey techniques is generally attributed to the large percentage of individuals who

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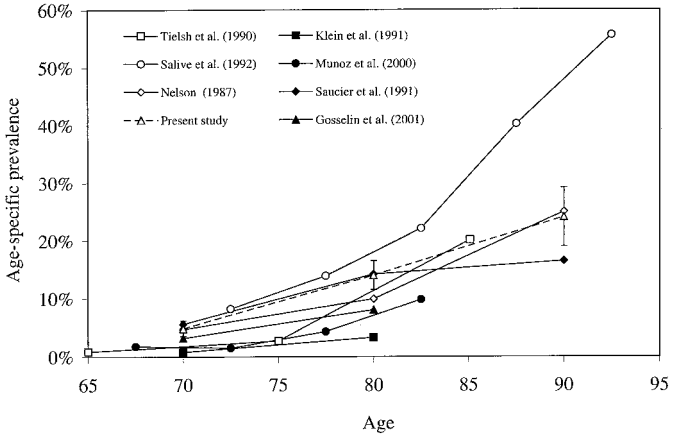


FIGURE 1.

Age-specific prevalence of visual impairment in elderly subjects from seven recent studies, using midpoints of the age ranges. The results of the present study are presented with a 95% confidence interval.

report visual disability because of inadequate optical correction, uncorrected presbyopia, or medically or surgically remediable conditions.[27–31] Large discrepancies between prevalence of visual impairment based on presenting and best-corrected VA were also observed in several population-based studies.[5, 19] Other investigators have shown that when visual impairment is defined on the basis of at least one questionnaire response, the prevalence of moderate visual impairment may be overestimated by a factor of two.[27, 28]

There was a higher prevalence of problems in reading ordinary newsprint than of problems in seeing the face of someone 12 feet away. This finding is consistent with other studies[3] and suggests a hierarchy in task difficulty levels: to be able to read ordinary newsprint, an individual has to present a VA of at least 6/15. On the other hand, both reading and face recognition are contrast dependent.[31] However, because faces subtend a rather larger visual angle than ordinary newsprint, face recognition might be considered a more robust task.[31] Consequently, individuals reporting both visual disabilities are probably more seriously affected than those reporting only one. This assertion is also corroborated by the fact that individuals with two positive responses use rehabilitation or nonprofit agencies services more frequently than those with only one positive response (Table 3). Overall, the proportion of subjects who used rehabilitation services in this study was low. Yet, we found that among individuals who use at least one service, many use more than one. The utilization of rehabilitation centers and nonprofit agencies that offer visual aids at no cost or low cost was particularly low. Although the proportion of individuals reporting the use of low-vision clinics in hospitals or private offices was higher, this may simply reflect the respondent's confusion of low-vision services with other ophthalmologic subspecialties because the utilization rate was identical among those with one and those with two positives responses.

When visual impairment was considered to be present if one response was positive (as proposed by Statistics Canada), we obtained a high NPV, indicating that a large proportion of individuals with a negative test are truly unimpaired. On the other hand, the PPV was low: only 21% of those with a positive test had moderate functional impairment or worse. In this sample, there was little difference in the prevalence of functional visual impairment between individuals with disability at distance and those with both near- and far-distance disability. The use of either the distance or the distance and near visual disability criteria resulted in an increase in the PPV of the questionnaire without markedly decreasing the NPV. However, in the case of moderate visual impairment, the estimated PPV was still quite low (57%).

If we assume that the estimates of positive and negative predictive value apply equally to those who refused the visual examination as to those who received it, we may use them to estimate the prevalence of true visual impairment in the population of Quebec aged at least 65 years in 1992. The adjusted prevalence of mild functional impairment and worse was not very different from the prevalence of reported visual impairment based on a criterion of at least one positive response on the questionnaire (9.5% versus 8.5%, respectively). Similarly, the adjusted prevalence of moderate visual impairment and worse did not differ much from the value obtained using a criterion of two positive responses (2.6% versus 3.5%).

The study had several limitations. First, the participation rate in the validation component was low. This may have been due to the long interval (approximately 6 months) before recontacting the subjects. Nonresponse bias (e.g., differential participation according to gender), therefore, may be an issue in this study. Other selection biases might be responsible for the unusual absence of differences in the levels of presenting acuity and best-corrected acuity among the examined subjects. Here, the low level of participation in the second stage of the study may have introduced a bias toward the enrollment of individuals with appropriate ophthalmic correction. Also, the relatively small number of subjects resulted in very wide confidence intervals for some of the estimates.

The question regarding the utilization of rehabilitation services was only asked of individuals with at least one visual disability complaint. It is highly unlikely that individuals who answered negatively to both questions on the visual screening questionnaire would use visual rehabilitation services. From the corrected utilization rate of rehabilitation services, the number of individuals using rehabilitation centers or nonprofit agencies in Quebec can be estimated to be around 5000 persons for 1992. This finding is consistent with registration numbers in the rehabilitation centers and the nonprofit agencies throughout Quebec at the time of this study.[10, 32]

In this study, we estimated the predictive values and misclassification rates of the HALS survey questions to be able to derive the prevalence of visually impairment and blindness. Our results may seem to be in contradiction with the results of other investigators.[29, 30] For Elliott et al.[29] the prevalence of low vision and blindness could not be inferred from the results of the HALS survey because they did not find a direct relationship between the HALS results and the number of registrants at Ontario low-vision clinics and the Canadian Institute for the Blind. We agree with these authors[29, 30] that questionnaire-based results overestimate the true prevalence of visual impairment. Nevertheless, our results suggest that with knowledge about the properties of the survey questions, the results of the questionnaire survey might be used to estimate the prevalence of visual impairment.

Another important finding of the present study is related to the

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utilization rate of services. Other studies corroborate the low utilization rates found in our study. In Great Britain, fewer than 25% of the fully eligible individuals among the general population are registered with the Royal National Institute for the Blind.[33] Low participation rates may be attributed to a variety of factors. First, there may be a lack of awareness in the population concerning the difference between pathologic visual decline and visual decline related to aging.[34] Seniors (as well as health care professionals) may have low expectations regarding the vision of elders,[35, 36] referrals may involve long waiting times,[36] both elders and professionals may have little knowledge about available services,[37, 38] and elderly persons may not adhere to the recommendations of their ophthalmologist.[39]

Rehabilitation centers were originally established to preserve the full independence of visually impaired individuals by helping them to cope with visual loss and by reducing their disability and handicap situations. In the elderly population, rehabilitation services may preserve independence and prevent the need for institutionalization. This may imply a better quality of life for these individuals and a reduced cost for society as a whole.[40] An unknown percentage of individuals who report visual disability who undergo surgical or medical treatment of their visual problem can be expected to no longer require rehabilitation services.[30] However, the low utilization of rehabilitation services observed in this study raises the need to address the respective role of general eye care practitioners, community-based health centers, and rehabilitation centers in the rehabilitative process of visually impaired elderly individuals.

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